Ponuel Mollah Swadesh

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ResearchGate | LinkedIn | GitHub | Portfolio

RESEARCH INTERESTS

Data-driven decision support systems; Applied machine learning; AI for health and social impact; ICT for development; Geospatial AI; Digital public services.

EDUCATION

BSc in Computer Science & Engineering **2021**
Brac University, Dhaka, Bangladesh
CGPA 3.71 / 4.00

Higher Secondary Certificate (Science) **2016**
Bangabandhu Govt. College, Khulna, Bangladesh
GPA 5.00 / 5.00

Secondary School Certificate (Science) **2014**
St. Joseph's High School, Khulna, Bangladesh
GPA 5.00 / 5.00

ACADEMIC ACHIEVEMENTS

- Graduated with High Distinction.
- Merit-based Scholarship, Brac University (2017 to 2021).
- Board Scholarship, Higher Secondary Certificate (2017 to 2020).
- Five-time Dean's List Awardee: Spring 2017, Summer 2019, Spring 2020, Fall 2020, Spring 2021.

PUBLICATIONS & RESEARCH

1. A. J. Ramisa, A. Hossain, S. M. I. Islam, **P. M. Swadesh**, M. T. Islam, M. A. Rahman, M. Z. Parvez, "Gene expression data classification and pattern analysis using a data-driven approach," 20th International Conference on Machine Learning and Cybernetics (ICMLC), Adelaide, Australia, 2021. (Published)

- **DOI:** 10.1109/icmlc54886.2021.9737248
- **Research area:** Applied machine-learning methods for high-dimensional gene-expression data, including feature selection, classification, pattern analysis, and model evaluation.

2. M. H. Islam, N. A. S. Jiku, S. Mandal, **P. M. Swadesh**, and A. Ghosh, "Beyond Single Transformers: A BanglaBERT-XLM-RoBERTa Stacking Framework for Code-Mixed Emotion Recognition," 2026 3rd IEEE International Conference on Computing, Applications and Systems (COMPAS). (Accepted)

- **Research area:** Natural Language Processing (NLP), Artificial Intelligence.

3. M. H. Islam, N. A. S. Jiku, S. Mandal, **P. M. Swadesh,** and A. Ghosh, "Automated Visual Inspection of Tilapia Fish Quality Using CNN and Transfer Learning," IEEE International Conference on Optics, Machine Learning and Emerging Technology (OMLET 2026). (Accepted)

- **Research area:** Artificial Intelligence and Computer Vision

4. **P. M. Swadesh,** A. Ghosh, K. Asif, S. Mandal, A. Dutta, S. A. Sworna, "An Explainable Hybrid Stacking Ensemble for One-Hour-Ahead Soil Moisture Forecasting Toward Cost-Aware Irrigation Decision Support," 14th International Conference on Electrical and Computer Engineering (ICECE 2026). (Under review)

- **Research area:** Explainable Artificial Intelligence (XAI), machine learning, time-series forecasting, soil-moisture prediction, and AI-driven irrigation decision support.

5. D Mondol, A. Ghosh, S. Mandal, J. Alam, P. Tanjin, A. Dutta, K. Asif, **P. M. Swadesh**, and R. Baidya, "Explainable Hybrid CNN-BiLSTM-Transformer Network Intrusion Detection with Unknown-Attack Rejection," 12th IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering 2026. (Accepted)

- **Research area:** Security and Information Assurance

6. Enhanced U-Net for Informal Settlement Detection in High-Resolution Satellite Imagery: Model Performance, Field Validation, and the Limits of Roofing-Material Proxies. (Working paper) | WaterAid Bangladesh

- **Research area:** Applied deep learning methods for very high-resolution satellite imagery analysis, field-based ground-truth validation, and geospatial dashboard deployment for urban informal settlement detection.

7. Chameleon Words in Cross-Domain Sentiment Analysis: An Explainable Framework with LLM-Based Style Transfer. (Working paper)

- **Research area:** Natural Language Processing (NLP), Artificial Intelligence.

PROFESSIONAL EXPERIENCE

ICT & Digital Development Officer **October 2025 to Present**

WaterAid Bangladesh

- Lead and coordinate digital systems, mobile applications, dashboards, and data workflows used by communities, municipalities, water utilities, and development partners.
- Translate public-service and service-delivery challenges into system requirements and practical digital solutions.
- Support the digitization of Chattogram WASA customer-service processes.
- Designed GIS-enabled municipal information workflows integrating spatial and administrative datasets to support sanitation planning.
- Support organizational adoption of AI, automation, data management, digital accessibility and information security.
- Supervise vendors, external developers, and ICT personnel, ensuring quality assurance, delivery timelines, technical documentation, and user-centered implementation.

Associate Officer – ICT & Digital Development **January 2024 to September 2025**

WaterAid Bangladesh

- Managed digital platforms addressing menstrual health, gender-based violence, public sanitation, customer services, water management and municipal information systems.
- Led the development and ongoing management of the INGO Forum Bangladesh website.
- Coordinated implementation of the Integrated Municipal Information System (IMIS) for Lalmonirhat Municipality.

Young Professional – ICT & Digital Development **January 2023 to December 2023**

WaterAid Bangladesh

- Developed and maintained the Ananya app v1.0, a mobile app for menstrual hygiene knowledge sharing among women and girls in underserved communities.
- Administered and customized the Drupal-based WaterAid Bangladesh official website, covering CMS configuration, maintenance and content workflows and MHM Platform website.
- Developed the Vendor Enlistment Platform, enhancing process automation and database-driven application design.

Junior IT Executive **January 2022 to December 2022**

FrogBID, Khulna, Bangladesh

- System setup and maintenance; data and asset management; end-user technical support; and vendor coordination.

Teaching experience **January 2019 to September 2021**

Freelance

- Peer mentor (BRAC University) and private tutor (Mathematics, Physics, ICT):); led exam-preparation study groups and mentored junior and Grade 8 students.

KEY PROFESSIONAL PROJECTS

Ananya Menstrual Health management App:

- Developed and maintained a mobile application designed to improve access to menstrual-health information for women and girls in underserved communities, reaching **45,000+** registered users.
- Supported features including period tracking, health-information access, hospital and helpline information, and an AI-enabled chatbot developed with program teams, medical reviewers, and implementation partners.
- Gained practical insight into challenges of digital health adoption, health-information access, user needs, privacy considerations, and equitable service delivery.

Chattogram Water Supply and Sewerage Authority (CWASA) Customer Service Application:

- Contributed to the design of a bilingual digital customer-service platform for billing, online payments, new-connection applications, complaints, notifications, and service-status tracking.
- Worked on translating utility service workflows into digital requirements that can improve visibility of customer requests, service bottlenecks, and operational response needs.
- Built experience relevant to future research on resource-constrained public-service systems, where service demand, infrastructure capacity, and response priorities must be balanced.

Municipal Sanitation & WASH Information Systems.

- Supported GIS-enabled municipal information systems and digital data-collection workflows for sanitation, public toilets, water quality, infrastructure, and service coverage, including coordinating a GIS drone survey and WASH situation assessment across Lalmonirhat and Paikgacha with SNV.
- Worked with program teams to organize field, administrative, and spatial data for monitoring and planning.
- Developed practical understanding of the data-quality, interoperability, capacity, and equity challenges that affect data-driven decision-making in low-resource public-service settings.

Toilet Conference 2025:

- Digital systems for an international sanitation conference with 500+ delegates from 28 nationalities, including website, registration portal, mobile app, digital payments, QR-based ticketing and check-in.

TECHNICAL SKILLS

Programming languages: Java, Kotlin, Python, JavaScript, Dart.
Machine learning and data science: PyTorch, scikit-learn, CatBoost.
Data analysis: NumPy, pandas
Geospatial computing: GeoPandas, QGIS, rasterio, GDAL, Leaflet.
Databases: MySQL, PostGIS, PostgreSQL, MongoDB, Firebase.
Mobile app and website development: Android Studio, Flutter, VS Studio, PHP, Next.js, Webflow, WordPress.
Tools and workflows: Git/GitHub, Microsoft 365, Power BI, Power Automate, Canva, Jira.

LANGUAGE PROFICIENCY

English: IELTS overall 7.0; Listening 7.5, Reading 8.0, Writing 6.5, Speaking 6.5. || **Bengali:** Native proficiency.

CERTIFICATIONS

- Al Fluency Framework & Foundations – Anthropic, 2026
- AI fluency for nonprofits – Anthropic, 2026
- Digital Data Protection – WaterAid, 2025.
- Introduction to Generative AI – Google, 2024.
- Google Analytics –Skillshop, 2024.
- Programming for Everybody (Getting Started with Python)- University of Michigan, Coursera, 2021
- Ethical Hacking and Web Security – eShikhon, 2021.

COMMUNITY & CO-CURRICULAR ENGAGEMENT

- **Digital Inclusion & Menstrual Health Support** — WaterAid Bangladesh (2023–Present): supported adolescent girls' and women's access to menstrual health information via mobile app and awareness activities.
- **Youth & Cultural Engagement** — BRAC University Cultural Club (2020–2021): Secretary of Performance, coordinating cultural programmes.
- **Youth & Community Program Support** — Bangladesh Catholic Students' Movement (2014–2017): supported youth seminars on leadership and community participation.